

Reproducing Axelrod's Evolution of Strategies in the Iterated Prisoner's Dilemma

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Abstract—An ongoing interest in studying how cooperation evolves has led to a replication of Robert Axelrod's unreleased third Iterated Prisoner's Dilemma computer tournament from the 1980s, featuring genetic algorithms. This study investigates whether genetic algorithms can evolve strategies on par with Tit for Tat, the winner of Axelrod's previous two tournaments, for an Iterated Prisoner's Dilemma game. Two experiments are conducted for a non-evolving environment with fixed strategies as opponents to analyze the influence of Tit for Tat in the evolution process. Three further experiments are conducted for a co-evolving environment with individuals pitted against each other, and explores how introducing an elitist genetic algorithm or parameter tuning affects performance. Measurements of effectiveness for the evolved strategies are based on the mean score of an individual relative to its peers, as well as the maximum obtainable score in one game. The results show genetic algorithm based evolved strategies scored highly, but not optimally within the population on average. The best evolved strategies in an average run give a comparable score to Tit for Tat, but often fail to outperform it. This work provides a close replication of Axelrod's strategy evolution experiments which were never released publicly, while also exploring possible improvements to the original implementation.

I. INTRODUCTION

The Iterated Prisoner's Dilemma (IPD) is an extension of the singular PD game theory concept, used to model how trust and strategy develops in real world fields, ranging from international relations to economic trade, or even biology and social behavior. The game involves two players who each must choose to either cooperate or defect at the same time. The goal is to accumulate not only a higher score than the opponent, but also the highest score possible, with the payoff structure as follows: (i) if both players cooperate, then the payoff for both players is a score of 3; (ii) a cooperation from one player and defection from the other yields a score of 5 for the defector and 0 for the cooperator; (iii) if both players defect, the payoff for both players is a score of 1. Unlike a singular PD where the optimal solution is to always defect, an IPD involves repeated interactions between the same players, which allows for strategies to develop cooperation as a viable outcome.

The submitted strategy Tit for Tat (TFT) won in Axelrod's first and second computer tournaments, proving that simple, nice, retaliatory strategies can outperform more complex or

exploitative ones, and also that stable cooperation is possible [1], [2]. While it answered which strategies performed well, the question of how such strategies could arise and persist among self-interested individuals remained. Axelrod turned to Genetic Algorithms (GAs) as a way to model the evolutionary process without human intervention [3]. This allowed for testing whether cooperation could emerge from strategies that were initially random, and if similar characteristics of niceness, reciprocity, or forgiveness from TFT would appear naturally. This replication focuses on using GAs to evolve effective strategies relative to the TFT strategy, other individuals in the same population, and the maximum score.

In prior work, Axelrod conducted two IPD computer tournaments. Although the original Fortran code has been lost, surviving reports have preserved the general methodology and results of the tournaments. For his first tournament, 14 strategies were submitted by economists, political scientists, mathematicians, and computer scientists, with RANDOM added in as the 15th strategy to serve as a baseline for comparison. Each strategy was autonomous, but could only base its decision on the previous round's outcome. In addition, no strategy knew how many rounds remained, which opponent it was facing, or the opponent's score. The tournament structure had every strategy play against every other strategy, including itself, with each match lasting exactly 200 rounds. The total score was simply the sum of all payoffs across all matches, with the rankings determined by the total points, disregarding the win/loss ratio. The second tournament followed a similar format, except 62 strategies were submitted with unlimited memory, and the number of rounds were randomized around 200. This was to prevent strategies from knowing when exactly the game ended, as the scenario would be reduced to a series of singular PD where the optimal solution would be to defect starting from the last round, chaining all the way to the very first round.

Prior results from the first and second tournaments where TFT won found that qualities of a successful strategy were:

- 1) Nice: not being the first to defect
- 2) Forgiving: can retaliate, but doesn't hold a grudge
- 3) Retaliatory: after an opponent defects, immediately respond with a defection
- 4) Clear: predictable pattern, able to establish trust

which many people describe as similar to the old “eye for an eye” philosophy[4]. In an IPD, no single best strategy exists. Instead, the strategy that ends up performing the best is reliant on the interactions with the opposing strategy. For example, in a tournament where TFT is surrounded by the DEFECTOR strategy which only defects, TFT will always perform the worst. Table 1 shows the worst performing nice strategy DAVIS outperforming the best performing nasty strategy GRASSKAMP in the first tournament.

TABLE I. FIRST TOURNAMENT SCORES

Player	Average Score
TIT FOR TAT	504
TIDEMAN & CHIERUZZI	500
NYDEGGER	486
GROFMAN	482
SHUBIK	481
STEIN & RAPOPORT	478
FRIEDMAN	473
DAVIS	472
GRASSKAMP	401
DOWNING	391
FELD	328
JOSS	304
TULLOK	301
(NAME WITHHELD)	282
RANDOM	276

This paper has new randomized strategies replacing the old method of accepting submissions for strategies from participants. While evolving human submitted strategies works, initializing random strategies have the benefit of controlled starting points and avoiding human bias, creating a much fairer environment as opposed to having intuition based strategies that carry common patterns or existing knowledge, leading to early domination. The non-evolving experiments consist of evolving strategies created from the GAs, which play against 8 strategies that do not change. These are hand picked from Axelrod’s first tournament, since the original documentation did not disclose the specific ones used. Exchanging TFT for FELD strategy allowed for examining whether the GAs were simply copying the well performing strategies it played against, or logically converging to similar behaviors over generations. Co-evolving experiments were conducted first with 20 individuals to determine if the patterns

from Axelrod’s paper could be reproduced without the specific seed for the initial randomized strategies. An elitist GA was then introduced, and parameters for population size and generations were increased to see if a GA with guaranteed process would have a better performance rather than a more probabilistic reproduction process offered from a standard GA. Through different environments, the results show that a GA can evolve strategies deemed effective relative to TFT of its peers. Note that the maximum score of 755 which comes from a full payoff of 5 points in each round for a total of 151 rounds in a match is impossible due to the presence of particularly nasty strategies, e.g., DEFECTOR. These evolved strategies are also not able to outperform TFT on average most likely because TFT already sits near the ecological optimum.

Remaining sections of this paper follow a specific structure. Section II reviews related work with IPDs and Axelrod’s previous tournaments or his GA based one. Section III explains the binary chromosome representation for the strategies, as well as the standard GA used. The next section describes the interactive user interface created to showcase the results from the replicated experiments, and provides a way for the user to play against a well performing evolved strategy. Following that, Section V presents the preliminary results of all five experiments and how effective the strategies are. Lastly, the conclusion summarizes the key points, along with constraints and future work.

II. RELATED WORK

While a significant amount of work has been done on the IPD, work in relation to Axelrod’s previous tournaments is fairly sparse. The most likely reasons are lost FORTRAN code from the 1980s, unreleased code, decades of research leading to interest in more complex social dilemma games, and using more advanced techniques such as Large Language Models (LLMs) instead of GAs. Only within the last month, were Knight *et al.* able to reproduce Axelrod’s second tournament by restoring and using modern compilers to compile and build an interface for calling the surviving fortran strategy functions without modification [5]. They were able to reproduce Axelrod’s main findings of TFT prevailing and successful qualities for good strategy performance, but relied on the open-source Axelrod-Python library, and encountered minor discrepancies such as bugs [6]. This paper also uses the Axelrod-Python library to call the representative strategies in the non-evolving environment, as well as to host the tournament game, while acknowledging that it is not Axelrod’s original implementation, and therefore may result in minor discrepancies as well.

A newer approach to tackling IPDs using LLMs was also published this year by Fontana *et al.* using Llama2, Llama3, and GPT3.5 to study their behavior [7]. These researchers concluded that the LLMs overall behave at least as cooperatively as humans, with forgiving and non-retaliatory rates varying across different models. Although this approach is not strictly related to Axelrod’s work with GAs, it highlights the use of more modern techniques used by researchers to tackle the IPD.

III. METHODOLOGY

Five total experiments are conducted, with two for the non-evolving environment, and three for the co-evolving environment. All experiments that used a standard GA for this work followed Axelrod's implementation closely, consisting of 20 initially randomized strategies and 50 generations, with each generation having 151 total rounds. 30 runs were conducted for each experiment instead of 40 in the original, with each run being randomly seeded. The game actions cooperate (C) and defection (D), along with the payoff values remained the same as in Axelrod's paper (CC = [3,3], CD = [0,5], DC = [5,0], DD = [1,1]) which depends on the outcome of the interactions between the two participants.

A. Encodings and Decodings

The 20 initial strategies are memory-3 deterministic strategies, each encoded as a 70 gene binary string. For each round, an action is chosen based on up to the last three interactions. If the last three rounds are identical and have occurred before, the strategy will produce the same exact action every time. This is possible because each gene in the chromosome is an action corresponding to an encoded lookup table where all possible 3 move histories are mapped. The 70 genes which make up a chromosome are from 64 different combinations of three previous moves, along with 6 genes which define the premise for both players, or the three hypothetical moves made before the actual first round of a match. The search space is much too large (2^{70}) to be feasible for an exhaustive search, thus the use of a GA. Since the encoding is direct, the decoding consists of a simple lookup operation during the game.

B. Genetic Operators

Fitness proportionate (Roulette Wheel) selection with scaling is used. Individuals with a mean score receive a 1.0 weight corresponding to one expected offspring. One standard deviation above the mean would receive a weight of 2.0. This prevents premature convergence by keeping selection pressure moderate. The Parents are selected twice per mating independently, so the same parent could also be selected multiple times for different matings. The probability of selection is proportional to the fitness weight, so better individuals have a higher chance, but even poor performing ones still retain some chance, which maintains diversity.

A single-point crossover can occur anywhere from bit 0 to bit 69 of a chromosome, which includes the premise. Since bits 0-63 represent the lookup table entries while bits 64-69 represent the premise, a crossover point around bit 64 could lead to them evolving independently. In each generation, 10 single-point crossovers occur, matching Axelrod's original implementation.

Uniform bit-flip mutation with a rate of 0.01 (1% per gene) is applied to every offspring after crossover. Since there are 70 genes in a chromosome, the expected flips is 0.7 per chromosome. A low mutation in this case helps to preserve good solutions while still allowing for exploration. This also follows the mutation setup in Axelrod's paper.

C. Fitness Evaluation

The goal for individuals in all of the matches is to accumulate the maximum amount of points. This however, depends on the opponent's strategy. A perfect exploitation of the opponent will yield $5 \times 151 = 755$ points, while mutual cooperation leads to 453 points, and mutual defection gives 151 points. The fitness is calculated by the evaluation function shown in Equation 1 at the end of each generation, when every individual has played against every other individual.

$$\text{Fitness} = \frac{\text{Total accumulated score}}{\text{Number of opponents}} \quad (1)$$

The Fitness described is actually a mean score rather than the score for an individual match or a total accumulated score, as it proves to be more useful when comparing effectiveness to its peers and strategies from Axelrod's prior tournaments.

Note that an individual's final mean fitness is the score obtained after calculating the mean score from 30 runs, where each run's score was an average score from 50 generations, and a score in each of the generations is the average from all the matches played by the individual. A final best fitness follows the same structure but for the best score recorded. Figure 1 shows the hierarchical aggregation structure used to compute individual final mean fitness.

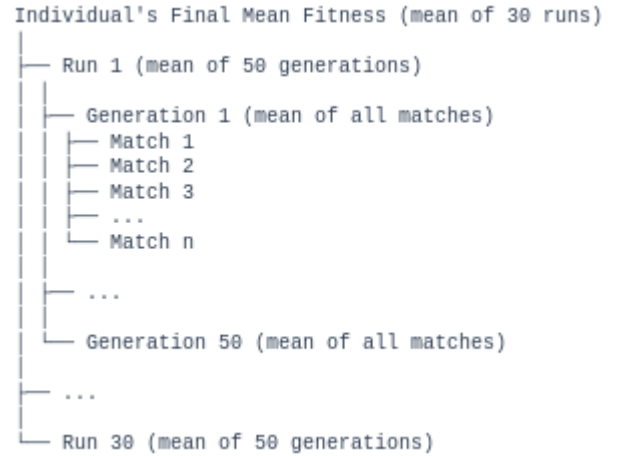


Fig. 1. Three-level hierarchical averaging structure for fitness evaluation

D. Non-evolving Environment

So far, almost everything discussed applies to all experiments. In the non-evolving environment, the 20 individuals are not playing against each other, but instead against a select 8 fixed strategies, or representatives. Two experiments were conducted, one with the presence of TFT, and another with the FELD strategy as a replacement to see what impact TFT had on the strategies that were being evolved. Since Axelrod did not disclose what specific representatives were used, a decision was made to personally select four nice (COOPERATOR, DAVIS, GROFMAN, TFT) and four nasty (DEFECTOR, GRASSKAMP, JOSS, RANDOM) strategies mostly from the first tournament, with FELD being a nasty replacement for TFT in the second experiment.

E. Co-evolving Environment

A co-evolving experiment does not use any fixed or submitted strategies. Instead, the 20 initially randomized individuals are pitted against each other, and must play everyone in the population each generation. After a generation, all 20 individuals can evolve before playing again, unlike in non-evolving where the opponents do not change. Of the three experiments done, one attempts to simply replicate Axelrod's results, while two others explore possible improvements by first exchanging the standard GA with an elitism GA, and then investigating whether increasing the population, generation, and number of elites can produce evolved strategies with a higher mean score than TFT or even the regular GA.

The second co-evolving experiment introduces an elitist GA, which ensures the highest scoring solutions found in one generation are preserved and carried over to the next generation without being changed by mutation or crossover. The number of elites is set to 2, or 10% of the population. Having more does not help, as too much elitism can cause a convergence to a suboptimal strategy, and the population is small so genetic diversity must be preserved. The third co-evolving experiment builds on top of the elitist GA by doubling the population and generations to 40 and 100 respectively, while also increasing the number of elites to 4.

Another way to assess effectiveness of evolved strategies is by looking at diversity, or the variety of different individuals in a population. The average Hamming distance shown in Equation 2 counts how many genes are different between chromosome pairs, and provides an average across all the pairs within a population.

$$D_{pop} = \frac{2}{N(N-1)} \sum_{i=1}^{N-1} \sum_{j=i+1}^N \sum_{k=1}^L 1(c_i[k] \neq c_j[k]) \quad (2)$$

For this equation, $\sum_{k=1}^L 1(c_i[k] \neq c_j[k])$ is one instance of the Hamming distance, while N is the population size, L is the chromosome length, and $\frac{2}{N(N-1)}$ is the number of unique pairs. High diversity means more exploration or finding new strategies while low diversity leads to more exploitation or refining strategies that are already good [8].

IV. INTERACTIVE INTERFACE & GAME

A contribution of this paper beyond reproducing Axelrod's unreleased third tournament is the web UI, which showcases results of how the GA performed across 30 runs for the non-evolving (TFT and without TFT), and co-evolving (with elitism) experiments. Statistics such as the representative strategy performance which details the mean score of the representatives (non-evolving only), as well as the average final best score, final average score, and best score achieved for the GA are listed. In addition, analysis for the best chromosome in each of the mentioned experiments are given such as the cooperation rate against every other opponent, its final premise, and the full chromosome which can easily be loaded into the game.

Note that the UI is designed using HTML, and does not require any external dependencies. All images were converted to base64, which allows for embedding directly into the code.

The game invites users to a one on one IDP tournament by either uploading a JSON or TXT file containing a valid chromosome, creating/pasting their own 70 bit chromosome, or using the load feature from the experiments on the side. The number of rounds has been defaulted to 10, but can be increased or decreased. A GA fitness gallery at the bottom shows graphs of mean scores and diversity (co-evolving only) for all five experiments conducted for this replication.

V. RESULTS AND DISCUSSION

Since individuals in the initial population are randomized, it is not reasonable to expect identical fitness in any generation or run. However, finding averages across runs with 30 random seeds can give a reasonable range for evaluation and comparison. The type of environment used has a direct impact on how a population performs throughout generations, as well as the strategies that dominate.

A. Non-evolving Results

The first experiment (α) and second experiment (β) deals with 20 randomized individuals playing against the 8 selected representatives, which were mentioned in a previous section. Table II below shows the mean score obtained across 30 runs for each of the representatives against the evolving individuals in both experiments.

TABLE II. REPRESENTATIVE STRATEGY PERFORMANCE α & β

Representative	Average Score (α)	Average Score (β)
TIT FOR TAT	442.6 (± 12.1)	N/A
JOSS	418.1 (± 22.4)	430.3 (± 22.4)
FELD	N/A	418.6 (± 25.6)
GROFMAN	356.7 (± 108.6)	351.5 (± 131.5)
GRASSKAMP	337.9 (± 65.9)	321.4 (± 74.0)
DAVIS	308.7 (± 116.2)	341.7 (± 120.8)
RANDOM	287.2 (± 74.5)	299.5 (± 76.7)
DEFECTOR	229.0 (± 69.6)	274.8 (± 101.9)
COOPERATOR	227.3 (± 178.9)	256.0 (± 197.2)

In the first experiment, TFT performs the best on average against the 20 individuals, which is an expected outcome, as it won Axelrod's two previous tournaments, and was only outperformed by the median individual for 11/40 runs. The small standard deviation of 12.1 also indicates that TFT is very consistent in establishing mutual cooperation. It does not exploit the individuals because TFT cannot initiate a defection, but only retaliate. On the other hand, the bottom performing representatives show much greater deviation,

hinting that they are prone to being taken advantage of by the nasty behaviors deployed by the evolved individuals. If an individual does not obtain any nasty strategies, then these representatives will achieve a much higher score than their recorded average.

For the related GA performance, the final best score recorded on average was 408.7 (± 7.4), while the final mean score on average was 392.7 (± 7.4), and the overall best score across all 30 runs was 423.6. Figure 2 shows both the average mean score and best score of individuals over 30 runs.

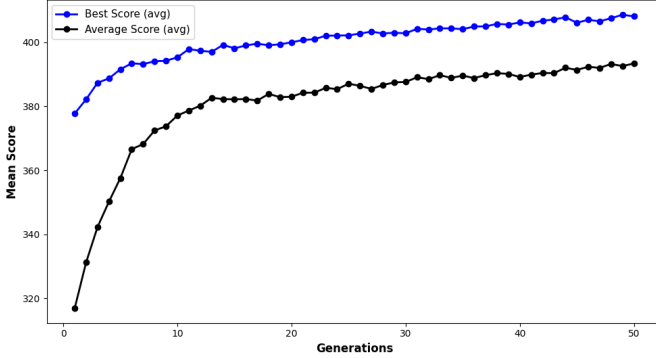


Fig. 2. Average mean and best performance of individuals with 30 random seeds for experiment 1. X-axis represents the number of generations and Y-axis shows the average fitness over 30 runs.

These scores are in line with Axelrod’s discovery that evolved individuals were effective relative to its peers, with the average individual ranking 3rd, but not being good enough to consistently outperform TFT. In this case, not even the best overall individual was able to outscore TFT. The best individual also had a nearly perfect cooperation rate with the top 4 representatives, roughly ~94~98%, while achieving a 39.4% overall cooperation rate. This statistic is not how many times mutual cooperation occurs, but simply how many times the best individual decides to perform the cooperation action on average. It suggests that this individual’s overall strategy is to maintain cooperation and not be the first to defect, since there were no representatives that with a 0.0% (0/151 moves) cooperation rate, but to also immediately retaliate on a defection. Furthermore, the ability to detect exploitable or very hostile strategies was acquired during evolution. It was able to realize that defecting was optimal against the COOPERATOR as it cooperated just 0.7% (1/151 moves) and 1.3% (2/151 moves) with DEFECTOR. In the case of RANDOM, a 7.9% (12/151 moves) meant the individual realized there was no way to maintain cooperation.

The second experiment saw the replacement of TFT with FELD as one of the fixed strategies, to see what potential impacts, if any TFT had on the evolution process. The FELD strategy is considered nasty, and does not exhibit the same behaviors. This meant that if evolved individuals still exhibited similar qualities as TFT, then it must have logically converged to those behaviors not from simply copying TFT, but because those behaviors work well in score accumulation within the given environment. JOSS performed the best with a relatively small deviation of 22.4, which is expected as it ranked second in the first experiment. The only difference

within the ranking is DAVIS outperforming GRASSKAMP. This swap is not very significant, especially when accounting for the large standard deviations of 74.0 and 120.8, so either strategy could perform better with luck, as it varies widely with runs.

The GA performance for the second experiment saw a final average best score of 394.9 (± 7.6), final average mean score of 378.4 (± 7.0), and an average overall best score of 409.8. Figure 3 visualizes the average mean score and best score for the individuals over 30 runs.

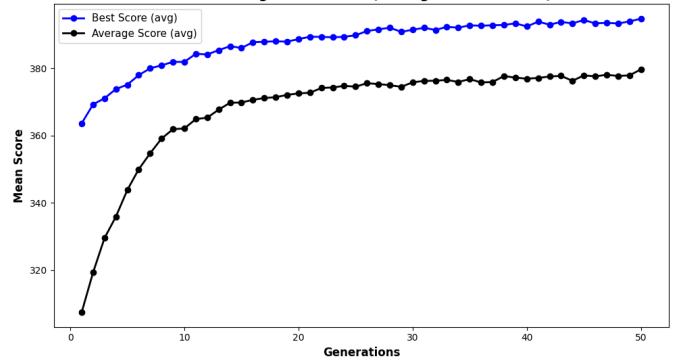


Fig. 3. Average mean and best performance of individuals with 30 random seeds for experiment 2. X-axis represents the number of generations and Y-axis shows the average fitness over 30 runs.

Similar to the performance found in experiment 1, both the average mean score and best score show improvements over generations. The mean score line is more parabolic at the start compared to the best score line, but both eventually taper off. This is expected, as the environment does not allow for infinite growth due to the presence of retaliation behavior and nasty strategies. Once again, the average individual scores highly but not the best, at 3rd. Even in a more hostile environment, a ~90-98% cooperation rate with the top 4 representatives was achieved, followed by a decent rate with FELD at 70.9% (107/151 moves). Interestingly, the overall rate increased to 45.9% (555/1208 moves), suggesting that the best chromosome for this experiment may be more forgiving. This might also explain why the average scores were all slightly lower, as the representatives may have had more opportunities to take advantage of the individual. Based on the 0.0% cooperation rates with COOPERATOR, DEFECTOR, and DAIS, some different conclusions can be reached. First, this evolved chromosome begins the match by defecting, most likely to test for the opponent’s strategy. It also has the ability to keep defecting indefinitely without triggering a truce behavior or something similar that attempts to establish cooperation. The best chromosome was also able to recognize the RANDOM strategy, as it only had a 11.3% (17/151) cooperation rate.

B. Co-evolving Results

Experiment 3 mirrors Axelrod’s original co-evolution study by removing the representative opponents, and having all 20 individuals of the population play against each other. The number of rounds per match, generations, and runs remain the same. This models real-life evolutionary dynamics, since organisms don’t evolve in isolation. Instead, they continuously

interact with each other, so their behaviors also influence one another.

Recorded performance for the GA were as follows: final best score on average was 428.5 (± 43.0), final average mean score was 407.5 (± 55.1), and the overall best score was 482.3. In addition, an averaged final diversity of 10.6 (± 2.1) was also calculated. Figure 4 shows the average mean score of individuals for 30 runs.

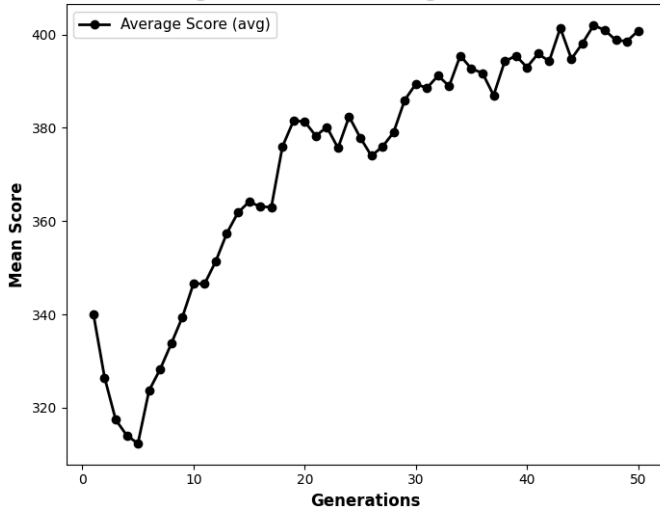


Fig. 4. Average mean performance of individuals with 30 random seeds for experiment 3. X-axis represents the number of generations and Y-axis shows the average fitness over 30 runs.

An improvement across all final scores compared to the previous experiments indicates that the population has learned to coordinate much more on mutually beneficial behavior, instead of exploiting fixed opponents. Moral behavior such as cooperation, fairness, and restraint can indeed emerge from just self-interest, but only when individuals respond to each other over time. In a non-evolving environment, this sort of feedback loop does not exist, as there are no consequences for how an individual behavior affects others long-term.

The evolution process is also different, as the average mean score does not experience a parabolic increase in the beginning. Instead, there is an initial drop in scores which accelerates briefly before a sharp reversal, followed by a higher climb, reconfirming the main pattern found from Axelrod's paper. This can be explained by some randomized individuals at the start having nasty behaviors, which are able to exploit others very well. The reproduction process therefore favors them, so more defection occurs, resulting in lower scores on average overall. The reversal is triggered when some individuals evolve towards conditional cooperation. Rather than cooperating more than defecting or at random intervals, the ability to establish trust and respond to hostile strategies with defection makes a strategy much harder to exploit. The overall cooperation rate for the best chromosome was 62.5% (1794/2869 moves). As it turns out, this combination is very similar to TFT, so the expectation is that final average scores are also similar.

Diversity tracking was also introduced for all co-evolution experiments as a way to study population patterns. It can help

explain whether the population is exploring, cycling, or collapsing, and whether cooperation is robust or fragile. A high average Hamming distance means a high sustained diversity, where no single strategy dominates others, and multiple niches like cooperators, exploiters, and punishers exist simultaneously. A low diversity can indicate mutual defection is common, or the population has converged to few strategies, believing it to be the best overall strategy. Figure 5 is a Hamming distance based diversity graph over 30 runs.

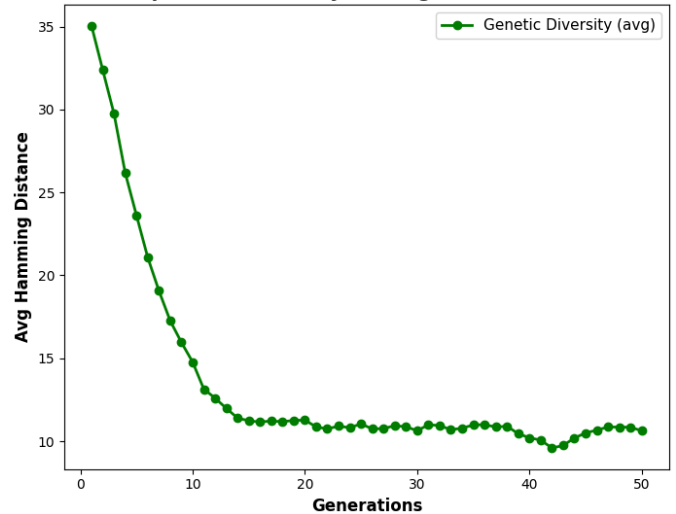


Fig. 5. Average Hamming distance of population over 30 random seeds for experiment 3. A decrease in distance means the variety of strategies in the population is shrinking.

The initial population diversity of 35 genes is the expected hamming distance for an initialized binary chromosome of length 70. Each gene position has a 50% chance of differing between any two random chromosome, so the expected distance is $L/2 = 35$ genes. The pattern shown of sharply decreasing before stabilizing out with minor fluctuations is also an intended outcome for diversity. As evolution proceeds, selection pressure causes the population to converge towards the high fitness solutions, which then decreases genetic diversity.

For experiment 4, the same setup applies, with an elitist GA configured with two elites replacing the standard GA. While elitism guarantees progress since the best score never decreases across generations, and should reach high-quality solutions in fewer generations, it is not without downsides. The algorithm focuses on refining good solutions rather than searching for new solutions, so better solutions in unexplored spaces might be missed. In the context of a co-evolving IPD, elitism makes it easy to lose good cooperative strategies, and acts as an anchor in the constantly changing fitness landscape. Too much elitism is a problem for dynamic opponents that need to adapt, and exploration is still needed to find the balance between cooperation and defection. The decision to use two elites was based on Goldberg's published work in 1989 [9].

The GA performance saw an average final best score of 414.0 (± 94.2), an average mean score of 366.5 (± 79.9), an averaged final diversity of 9.0 (± 2.9), and an overall best score

of 636.4. Figure 6 gives a visual representation for the average mean score of individuals over 30 runs.

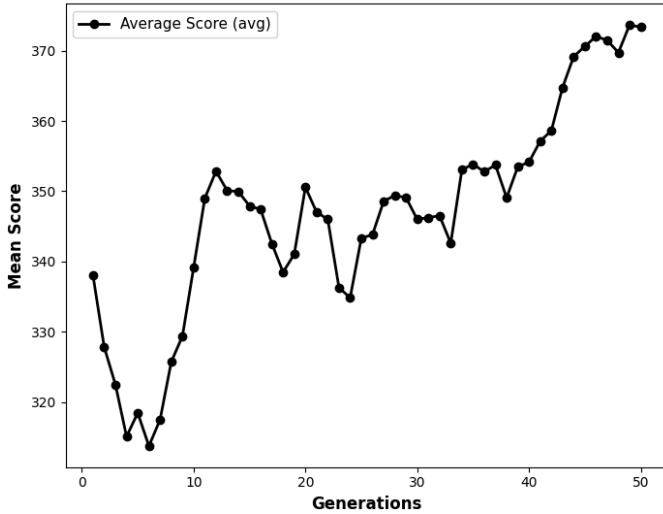


Fig. 6. Average mean performance of individuals with 30 random seeds for an elitist GA experiment. X-axis is the number of generations, while the Y-axis represents the average fitness over 30 runs.

The same pattern described for mean score in the previous experiment is seen here. After a reversal has happened and the mean score reaches a new peak at generation 12, notable fluctuations occur until around generation 40, where the average score consistently increases again. A reasoning for this might be due to a difference in quality between the elite individuals who are optimized for the previous generation's competitive environment, and the new offspring which are adapting to the current, evolving one. This gap, combined with the dynamic co-evolutionary fitness landscapes produces those oscillations. Towards the end however, genetic diversity decreases and offspring quality approaches elite quality. Interestingly, the average final scores were lower.

It seems to be counterintuitive, but a standard GA allows for all individuals to adapt to the current meta, and can abandon unsuccessful strategies, as there is no mandatory preservation of them. The standard GA can lose the best solution, while the elitist GA retains it, given by the best score of 636.4 compared to 482.3 from experiment 3. The cooperation rate for this best chromosome was 11.2%, suggesting it contained evolved strategies which were much less trusting. The diversity score was a bit lower, but overall negligible, as the standard deviation value was larger than the difference. When plotted, the average Hamming distance over generations for 30 runs was nearly identical to Figure 5.

Lastly, experiment 5 was built off of the implementation from experiment 4. The same elitist GA is used, with the parameters adjusted to see what effect, if any, it had on the performance. The initial randomized population was doubled from 20 to 40 individuals, the number of generations was doubled from 50 to 100, and the elitism count was increased from 2 to 4. Other parameters such as the mutation rate, number of matings, game rounds, and number of runs were unchanged, since the goal was to improve on average scores, not deviate from the original tournament design.

Statistics for the GA of this experiment were: 398.5 (± 83.2) for the average final best score, 378.8 (± 80.2) for the average mean score, 12.8 (± 3.0) for the average final diversity, and an overall best score of 514.9. Figure 7 shows the average mean score of the new population over 30 runs.

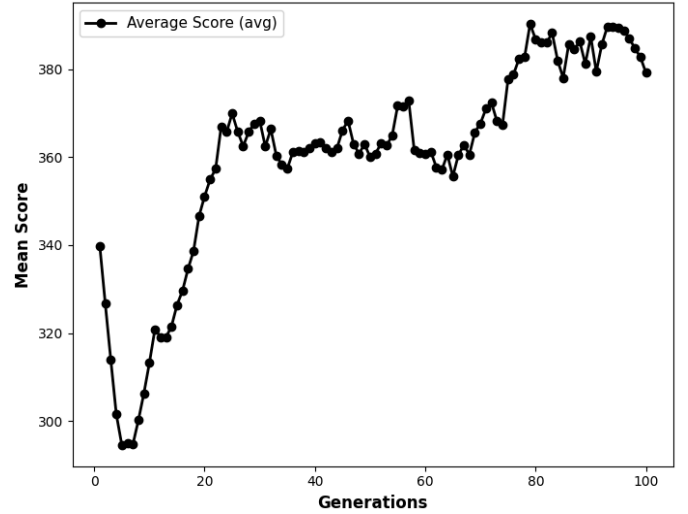


Fig. 7. The average mean performance with double the individuals and generations over 30 runs. X-axis is the number of generations, while Y-axis is the average fitness over the 30 runs.

As expected, the same pattern emerges. The average final scores were on par with the elitist GA from the last experiment. The diversity was a bit higher than the other co-evolution experiments, but not notable enough to draw new conclusions, as an identical decreasing pattern prevailed. One difference however, was the calculated 63.2% cooperation rate with the population, a stark contrast to the nastier and less forgiving final strategy evolved from experiment 4. In fact, this particular chromosome's cooperation rate is very close to the co-evolution standard GA experiment that had a 62.5% rate, which might indicate an independent convergence to very similar final strategies.

VI. CONCLUSION

This development project focuses on reproducing Axelrod's unreleased third tournament from the 1980s involving the use of genetic algorithms to investigate whether effective strategies could be naturally evolved. Five total experiments were conducted, with two in the non-evolving environment, and a further three with co-evolution. An analysis of the results concluded that evolved strategies were effective relative to its peers in the population, exhibiting conditional cooperation behavior with the presence and absence of Tit for Tat. The GA was confirmed to be logically arriving at similar strategies rather than simply copying strategies that it scored high against. It was, however, not able to outperform Tit for Tat.

For the co-evolution experiments, a straight replication of Axelrod's paper was first conducted, confirming the main pattern observed for average performance across generations, which was an initial decrease in cooperation due to exploiters dominating the population, followed by an eventual reversal

and higher peak in mean score once strategies evolved to be establish and maintain trust with the opponents. The other two experiments attempted to make slight improvements in average scoring by introducing an elitist genetic algorithm for preserving best scorers within one generation, as well as increasing parameters such as the number of individuals in the population and total generations per run. Preliminary results from the collected data showed that these changes still led to a respectable performance, albeit slightly worse than with a standard genetic algorithm. Part of this work's contribution included the creation of an interactive interface, allowing users to play against the evolved strategies from the experiments, or their own.

The main constraints with this project was the lack of documentation provided by Axelrod, as the experiment was never publicly released. The only code available were surviving fragments of strategies from the prior two computer tournaments. The community-developed Axelrod-Python library was used instead to host the tournaments and call the representative strategies, as Axelrod's original implementation was not in a modern language. Also, identical results cannot be achieved, due to the randomization of genes in the chromosome when being initialized, an intentional act meant to model natural evolution. Future work might involve using the three new methods for competitive co-evolution, as outlined in Rosin and Belew's research to improve on Axelrod's experiments [10]. Instead of genetic algorithms, more modern approaches such as Large Language Models could be used to evolve strategies and compare effectiveness. Lastly, applying genetic algorithms to other social dilemma games like Tragedy of the Commons can be an interesting way to study cooperation emergence, maintenance, and evolution.

ACKNOWLEDGMENT

Knowledge from this project was based on information obtained from the Evolutionary Computing course lectures and resources, along with guidance under Dr. Sushil J. Louis from the University of Nevada, Reno.

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PROJECT SOURCE CODE & DEMO LINK

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